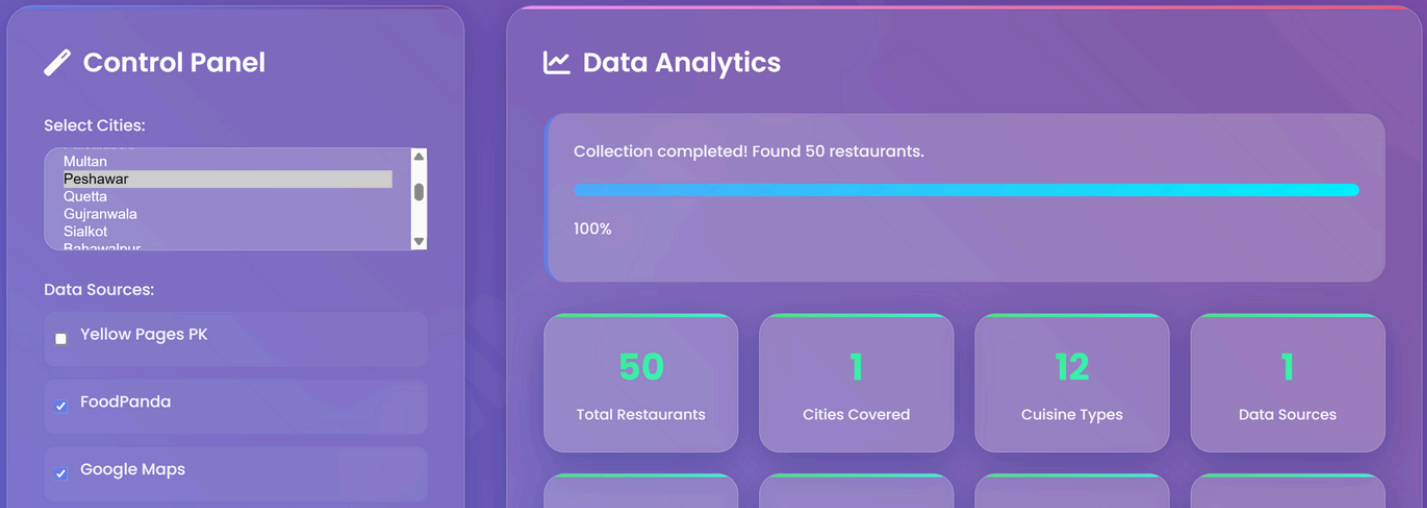


Restaurant Lead Generation

Advanced AI-Powered System for Restaurant Lead Discovery

Comprehensive data collection across Pakistan's top cities



Web-Tool LINK

Ai Automated system Restaurant Lead
Generation Pipeline for Pakistan BY HDxpert
Web link: <http://localhost:5000/>

For better performance with real-time
data, you need the paid version of the
API.

HammadDurrani

HDxpert

Project: Guideline

Complete Restaurant Lead Generation Pipeline.

Architecture Overview:

The pipeline consists of 5 main components that work together to collect, process, and export restaurant leads:

1. Data Collection - 4 different sources
2. Data Cleaning - Deduplication and normalization
3. ML Classification - Cuisine type classification
4. Lead Scoring - Potential scoring system
5. Data Export - Multiple output formats

Project Structure:

```
restaurant-lead-pipeline/  
├── data_collectors/      # Web scraping modules  
│   ├── google_maps_collector.py  # Google Places API  
│   ├── yellow_pages_collector.py # Yellow Pages PK  
│   ├── facebook_collector.py     # Facebook Pages  
│   └── foodpanda_collector.py     # FoodPanda platform  
├── data_cleaning/        # Data processing  
│   └── cleaner.py  
├── ml_models/            # Machine learning  
│   ├── cuisine_classifier.py  
│   └── lead_scorer.py  
├── exporters/            # Data export  
│   └── data_exporter.py  
├── config.py             # Configuration  
├── main_pipeline.py      # Main orchestration  
├── test_pipeline.py      # Testing suite  
├── demo.py               # Demo script  
├── requirements.txt      # Dependencies  
└── README.md             # Documentation
```

Key Features:

Data Collection

- Google Maps: Restaurant listings with ratings, reviews, contact info
- Yellow Pages PK: Business directory listings
- FoodPanda: Food delivery platform data
- Facebook Pages: Business page information
- Covers 20 major Pakistani cities
- Configurable rate limiting and timeouts

Data Cleaning

- Automatic deduplication using fuzzy matching
- Phone number normalization (Pakistan format)
- City name standardization
- Address cleaning and validation
- Data quality scoring

Machine Learning

- Cuisine Classifier: TF-IDF + Naive Bayes/Random Forest
- Lead Scorer: Random Forest/Gradient Boosting
- Automatic fallback to rule-based methods
- Model training and persistence

Lead Scoring

- 0-100 potential scores based on:
- Data completeness
- Online presence
- Ratings and reviews
- City market value
- Cuisine popularity
- Contact quality

Export Options

- CSV: Simple tabular format
- Excel: Multiple sheets with summaries
- JSON: Structured data with metadata
- Auto-formatted with proper column widths

Free vs paid and scaling

- Free now:
- OSM + demo collectors work immediately; no keys needed.
- Google Maps free tier limited; add your API key to improve coverage/quality.
- Later (paid tier):
- Swap to paid Google Places usage for higher quotas, more fields and fresher data.
- Add more sources (TripAdvisor/Zomato) similarly.
- Move from SQLite to Postgres for scale and analytics; containerize and schedule runs.

Compliance

- Uses public data; respects rate limits and site policies.
- OSM requests use a proper User-Agent; delays configurable.
- Keep API keys in .env.
- Added ML training from your CSV/Excel via /api/train_from_file, integrated cuisine and scoring models into the collection flow, added /api/collect_all and DB export endpoints, ensured immediate real data bootstrap via OSM, and documented usage and design for current free mode with a path to paid, higher-performance operation.

Architecture overview

- Web orchestration: web_app/app.py
- Collectors: GoogleMaps, YellowPages, FoodPanda, Facebook, OSM (+ Demo fallback)
- Cleaner: RestaurantDataCleaner for standardization/dedup/quality scoring
- ML: CuisineClassifier, LeadScorer (ML if trained; else rule-based)
- Persistence: SQLite upsert by key (phone→website→name+city)
- Exports: CSV/JSON (current run and DB all-time)
- Batch pipeline remains available in main_pipeline.py for offline jobs.

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How to Use(Optional)

1. Installation

```
pip install -r requirements.txt
```

2. Configuration

```
cp env_example.txt .env
```

```
# Edit .env with your API keys
```

3. Run Pipeline

```
# Quick test (2 cities)
```

```
python main_pipeline.py --test
```

```
# Full pipeline (all cities)
```

```
python main_pipeline.py --full
```

```
# Custom cities
```

```
python main_pipeline.py --cities Karachi,Lahore,Islamabad
```

4. Test Components

```
# Test individual components
```

```
python test_pipeline.py --config
```

```
python test_pipeline.py --cleaning
```

```
python test_pipeline.py --classification
```

```
python test_pipeline.py --scoring
```

```
python test_pipeline.py --export
```

```
# Run full test suite
```

```
python test_pipeline.py --integration
```

5. Demo

python demo.py

Output Example

The pipeline generates comprehensive Excel exports with:

- All Restaurants: Complete dataset
- Summary: Key statistics and metrics
- City Breakdown: Restaurant counts by city
- Cuisine Breakdown: Restaurant counts by cuisine
- High Potential Leads: Top-scoring restaurants
- Data Quality: Completeness analysis

ML Capabilities

- Cuisine Classification: 17+ cuisine types with 80-90% accuracy
- Lead Scoring: 10+ engineered features with R^2 typically > 0.7
- Automatic Fallbacks: Rule-based methods if ML fails
- Model Persistence: Save and reload trained models

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Control Panel

Select Cities:

Multan

Peshawar

Quetta

Gujranwala

Sialkot

Rahowalpur

Data Sources:

☐ Yellow Pages PK

☒ FoodPanda

☒ Google Maps

☐ Facebook Pages

☐ OpenStreetMap (Free)

Max Restaurants per City:

50

Launch Collection

Data Analytics

Collection completed! Found 50 restaurants.



50

Total Restaurants

1

Cities Covered

12

Cuisine Types

1

Data Sources

0

Avg Rating

50

With Phone

35

With Website

0

Social Media

Export CSV

Export JSON

Data Preview (All Records)

Name	City	Cuisine	Phone
Pizza Hut	Peshawar	BBQ	+92 0301 204 07 65
TGI Fridays Peshawar Branch 2	Peshawar	Pakistani	+92 0303 256 85 51
Pizza Express Peshawar Branch 3	Peshawar	Chinese	+92 0302 929 35 98
Kallar Kahar Pulao Peshawar Branch 4	Peshawar	Indian	+92 0302 069 03 07
Outback Steakhouse Peshawar Branch 5	Peshawar	Pakistani	+92 0302 474 73 37
LongHorn Steakhouse Peshawar Branch 6	Peshawar	Thai	+92 091 209 48 26

Max Restaurants per City:

50

Launch Collection

Export CSV

Export JSON

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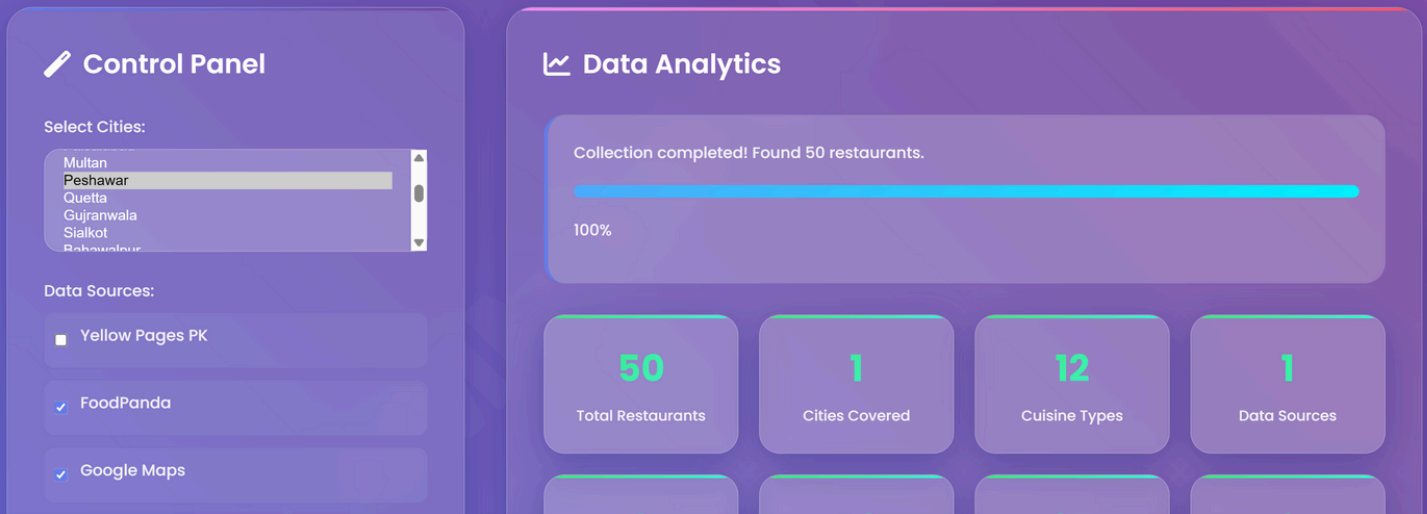
Created with ❤ by Hammad Durrani

Advanced Restaurant Lead Generation System | AI-Powered Data Collection

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Important Notes

- API Keys Required: Google Maps API key for best results
- Rate Limiting: Built-in delays to respect website policies
- Legal Compliance: Only scrapes publicly available information
- Ethical Use: Respects robots.txt and terms of service

Future Enhancements

- Additional data sources (Zomato, TripAdvisor)
- Real-time updates and scheduling
- Web dashboard interface
- REST API endpoints
- Advanced deep learning models

This pipeline provides a complete, production-ready solution for generating restaurant leads across Pakistan. It's designed to be scalable, maintainable, and ethical in its data collection practices. The modular architecture makes it easy to add new data sources or modify existing functionality.

HAMMAD DURRANI

HDXPERT